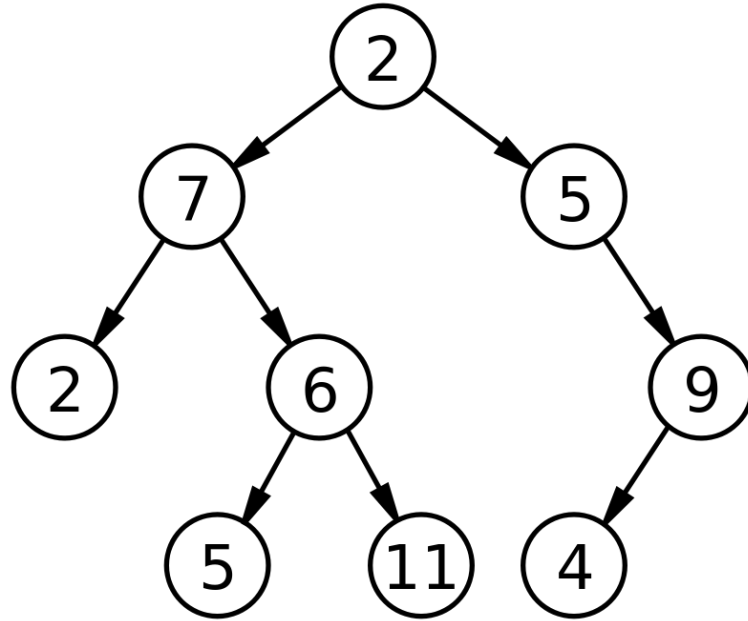

Data Structures

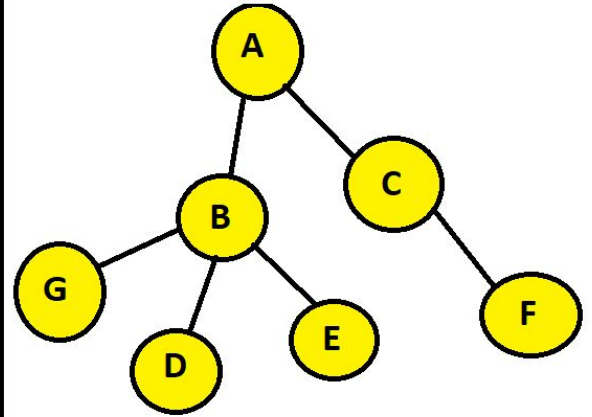
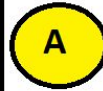
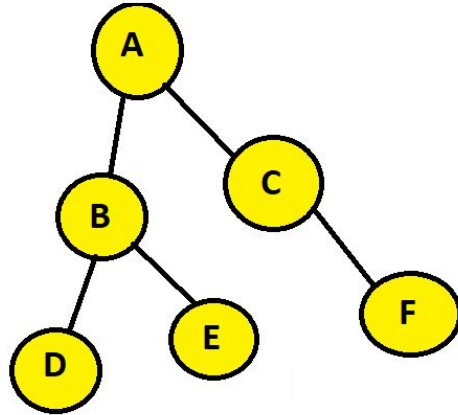
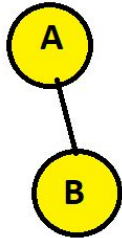
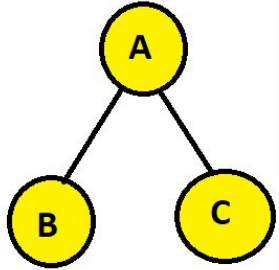
Binary Tree

Slides taken from Prantik Paul [PKP]

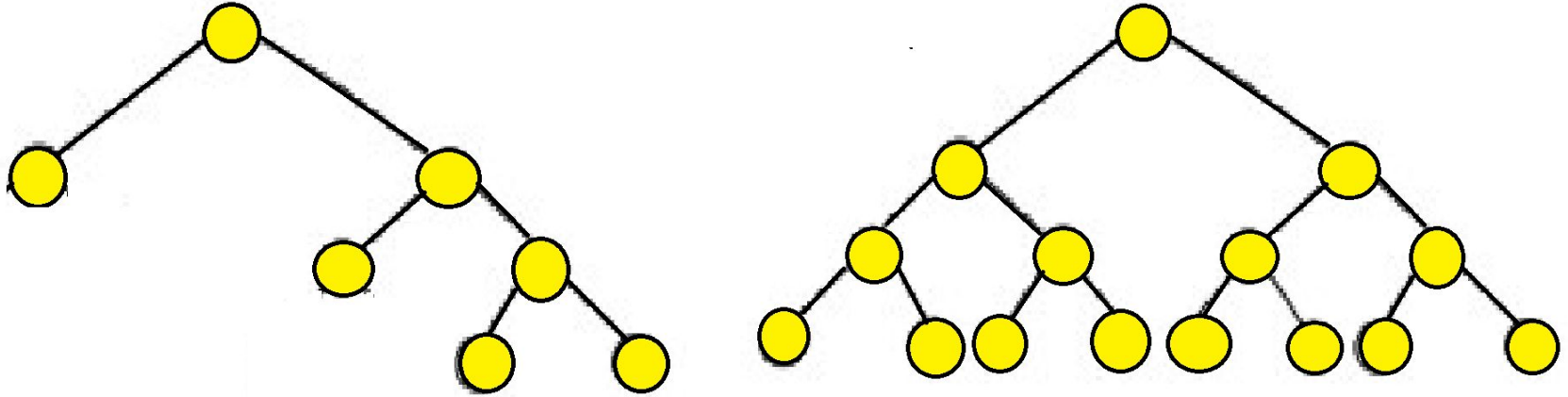
Binary Trees



Binary Trees

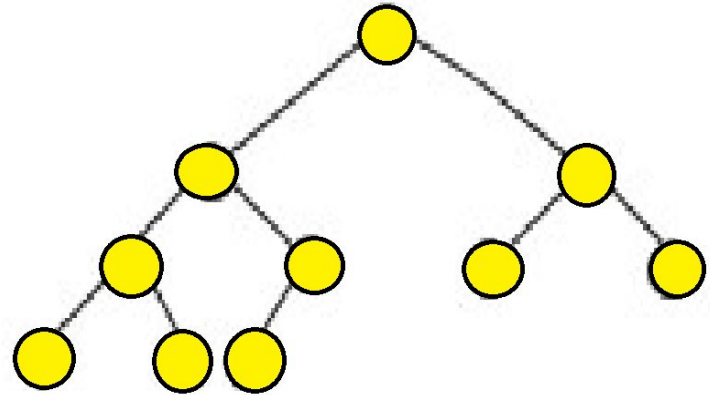
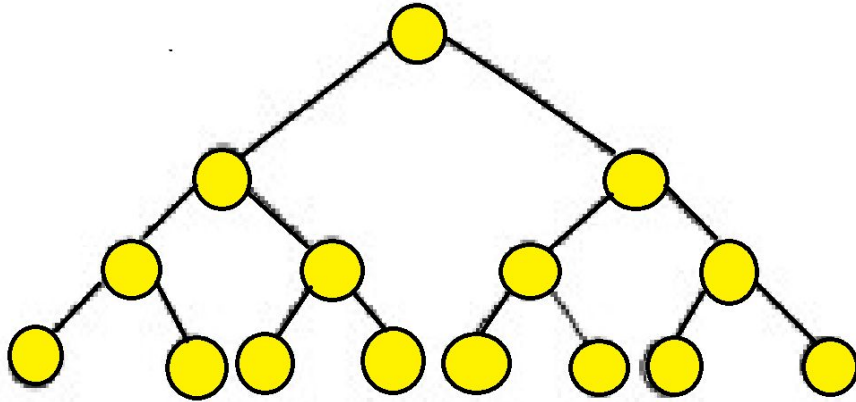


Binary Tree - Full/Strict Binary Tree



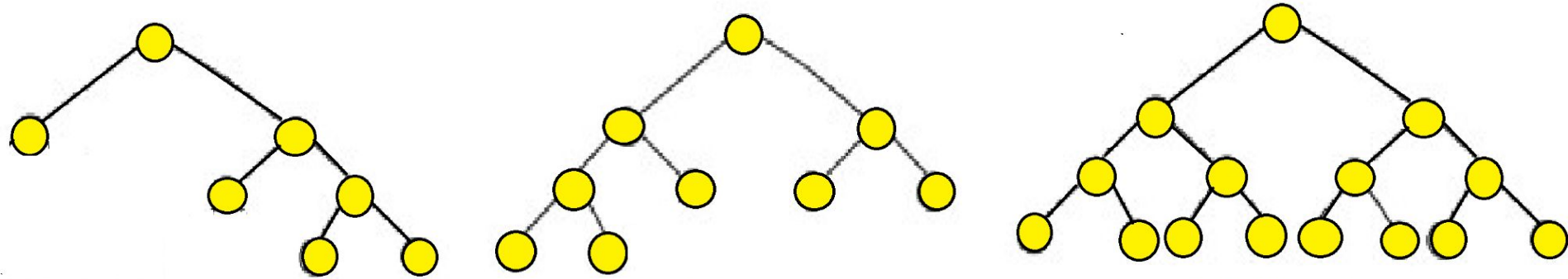
No of leaf nodes = no of internal nodes + 1

Binary Tree - Complete Binary Tree



All levels filled starting from LEFT

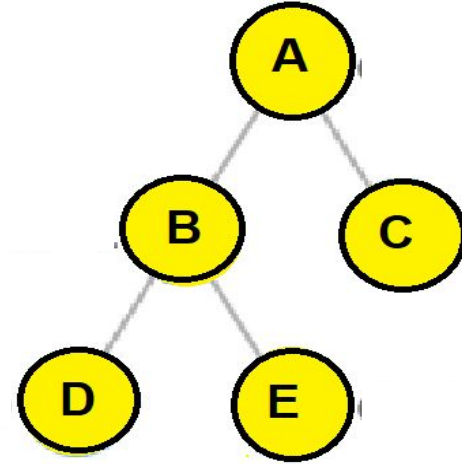
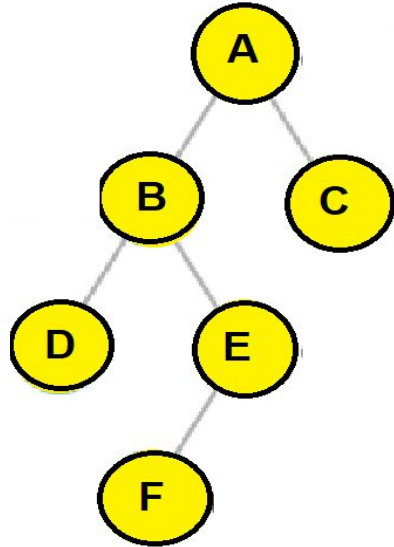
Binary Tree - Perfect Binary Tree



All Internal Nodes have 2 children

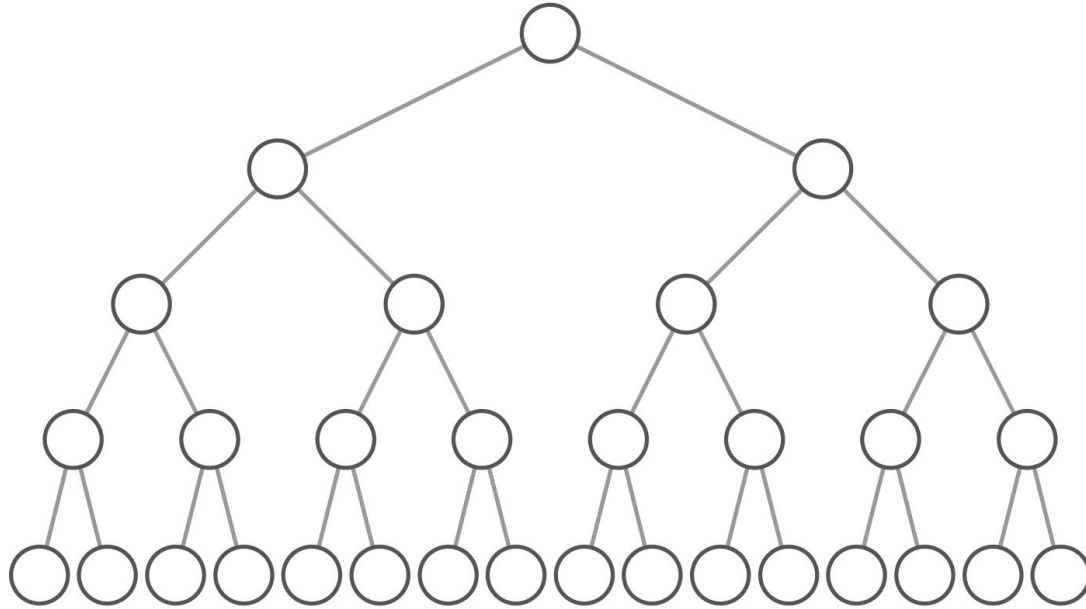
All Leaf Nodes at same Level

Binary Tree - Balanced Binary Tree



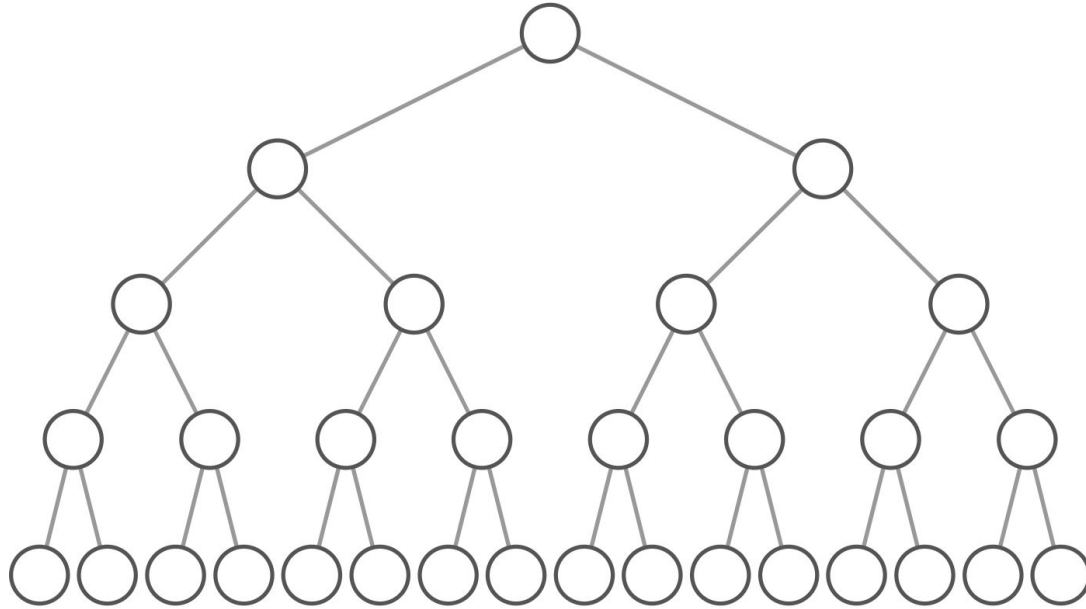
$$| \text{height}(\text{left}) - \text{height}(\text{right}) | \leq 1$$

Binary Trees - Characteristics



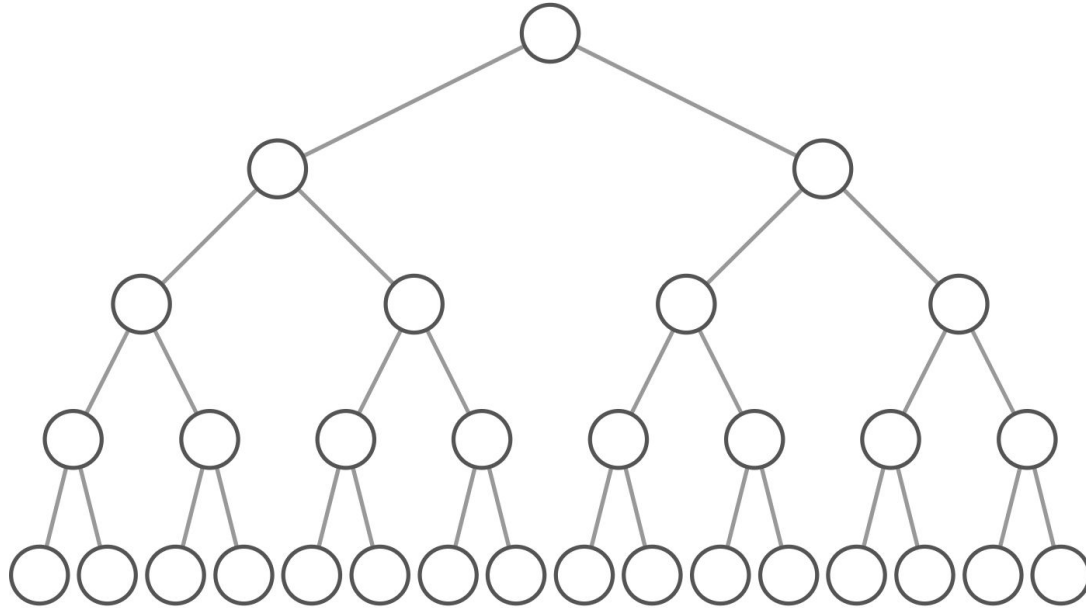
The maximum number of nodes at level i is:

Binary Trees - Characteristics



The maximum number of nodes possible in a binary tree of height 'h' is:

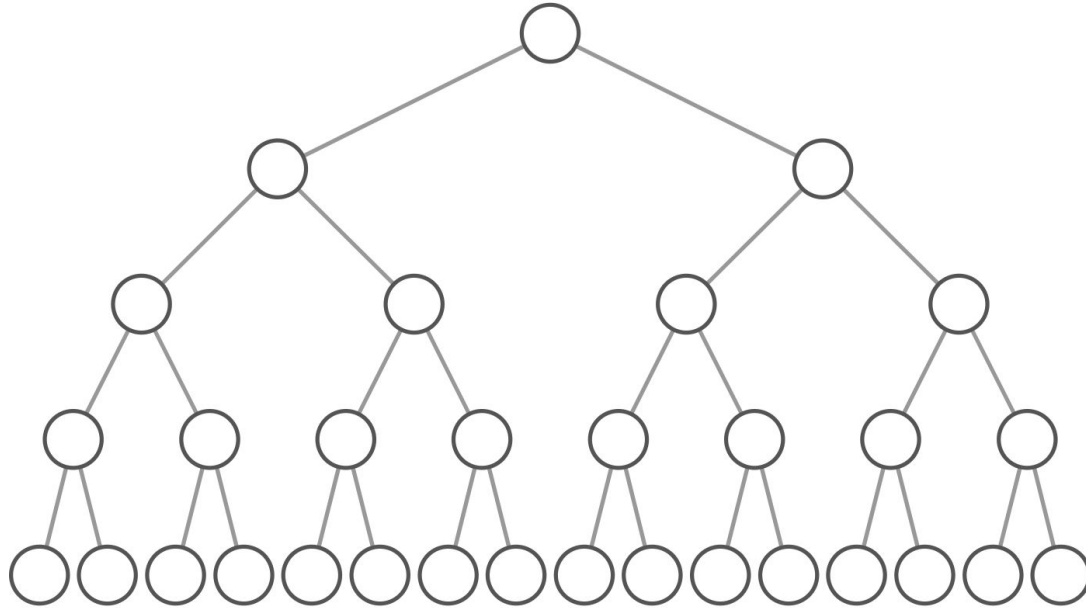
Binary Trees - Characteristics



Number of internal nodes : n

Number of external nodes :

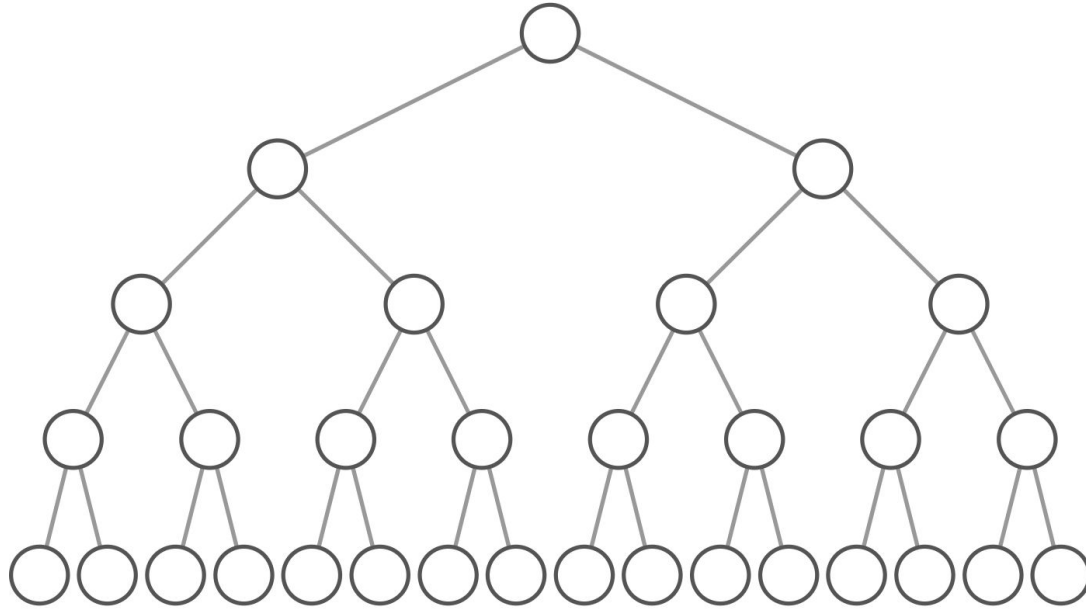
Binary Trees - Characteristics



Number of internal nodes : n

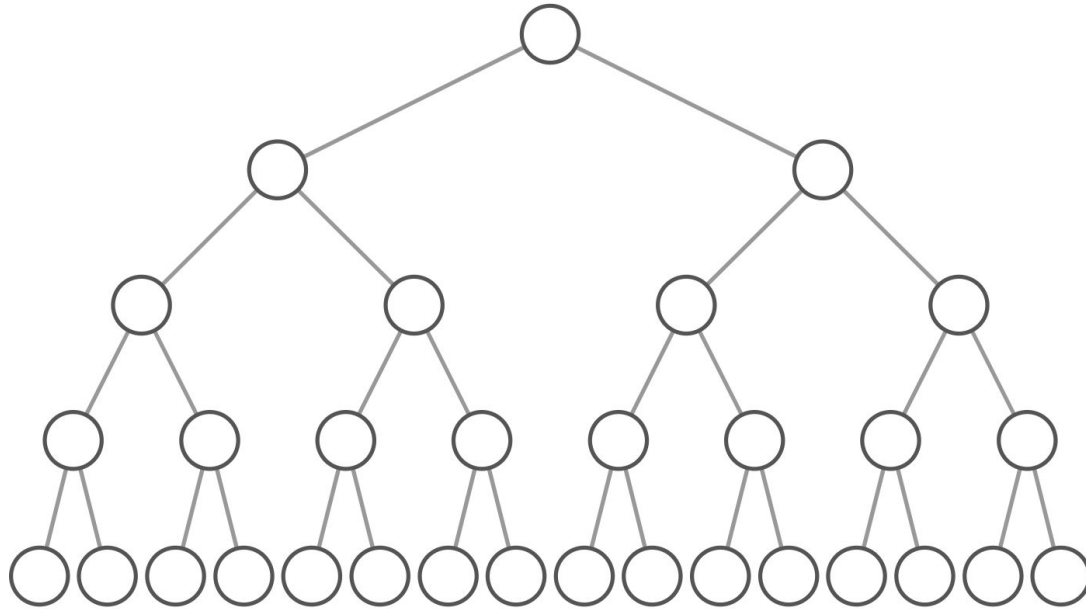
Number of internal edges :

Binary Trees - Characteristics



Number of internal nodes : n
Number of external edges :

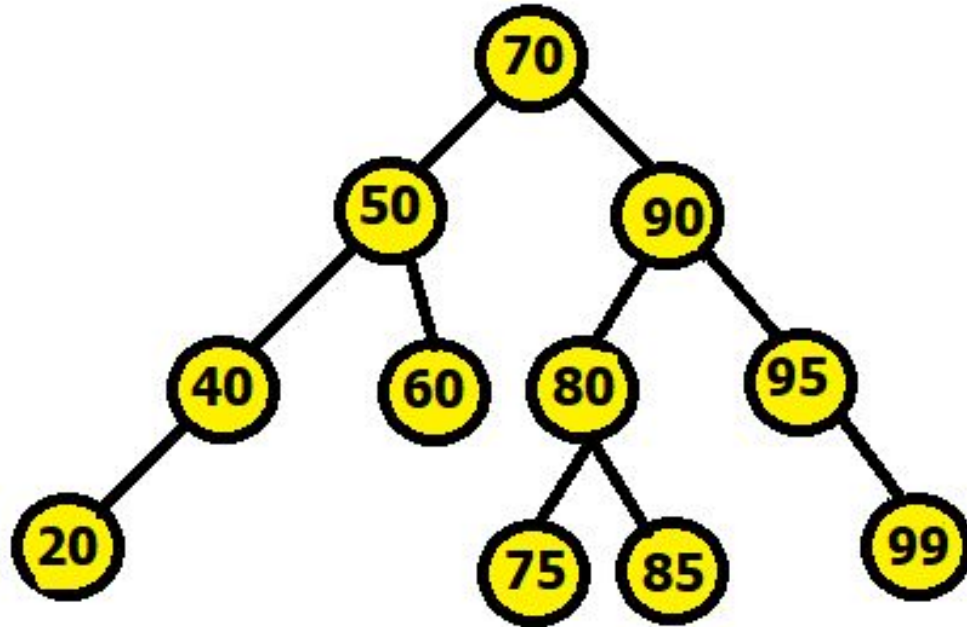
Binary Trees - Characteristics



Number of internal nodes : n

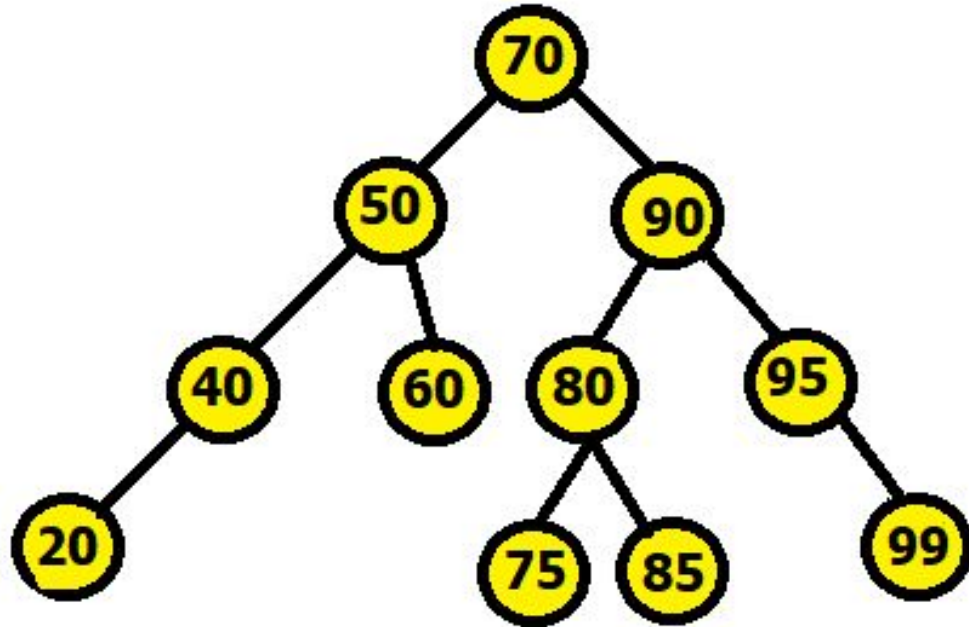
Number of edges :

Binary Trees - Traversal (Pre-order)



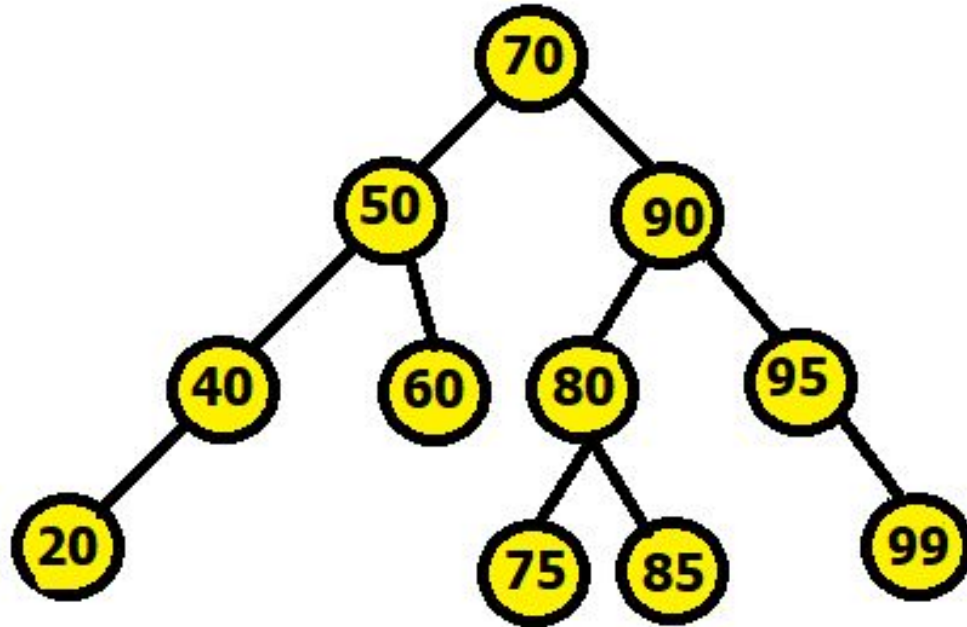
Pre-Order: 70, 50, 40, 20, 60, 90, 80, 75, 85, 95, 99

Binary Trees - Traversal (In-order)



In-Order: 20, 40, 50, 60, 70, 75, 80, 85, 90, 95, 99

Binary Trees - Traversal (Post-order)



Post-Order: 20, 40, 60, 50, 75, 85, 80, 99, 95, 90, 70

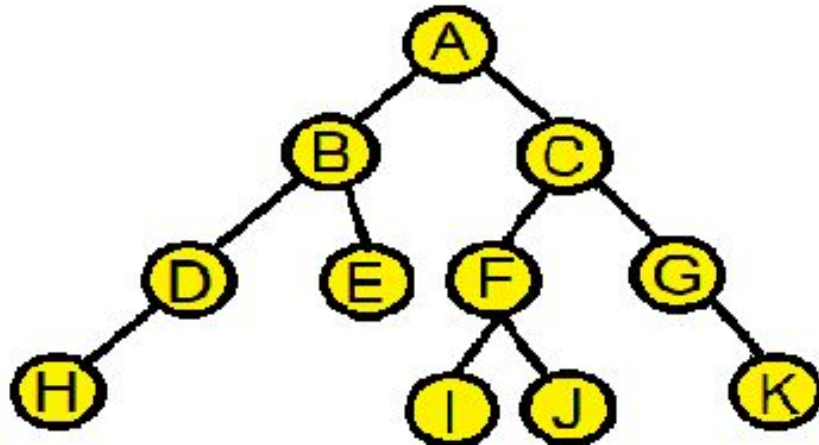
Binary Trees - Array Representation

If the **height** of the binary tree is **h** , An array of maximum **2^{h+1}** length is required

The **root** is placed at **index 1**

Any node that is placed at index **i** , will have its **left child** placed at **$2i$** and its **right child** at **$2i+1$**

Binary Trees - Array Representation



None	A	B	C	D	E	F	G	H	None	None	None	I	J	None	K
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Binary Trees - Tree Node

```
class Node:  
    def __init__(self, elem):  
        self.elem = elem  
        self.left = self.right = None
```

Binary Trees - Build Tree From Array

```
def tree_construction(arr, i, n):  
    root = None  
    if i < n:  
        if (arr[i] != None):  
            root = Node(arr[i])  
            root.left = tree_construction(arr, 2 * i, n) # insert left child  
            root.right = tree_construction(arr, 2 * i + 1, n) # insert right child  
    return root
```

Binary Trees - Preorder Traversal

```
def pre_order(root):  
    if root != None:  
        print(root.elem, end = " ")  
        pre_order(root.left)  
        pre_order(root.right)
```

Binary Trees - Inorder Traversal

Left → print(root.elem) → Right

Do yourself

Binary Trees - Postorder Traversal

Left → Right → print(root.elem)

Do yourself

Binary Trees - Build Array From Tree

```
array_rep = [None] * 16

def array_construction(n,i):
    if n==None:
        return None
    else:
        array_rep[i]= n.elem
        array_construction(n.left, 2*i)
        array_construction(n.right, 2*i+1)

array_construction(root,1)
print(array_rep)
```


Binary Trees - Count Nodes of Tree

```
def count(root):  
    if root == None:  
        return 0  
    else:  
        return 1 + count(root.left) + count(root.right)
```

Binary Trees - Find Level of a Node

```
def get_level(node, node_level, elem):
    if (node == None):
        return 0
    if (node.elem == elem):
        return node_level
    downlevel = get_level(node.left, node_level + 1, elem)
    if (downlevel != 0):
        return downlevel
    downlevel = get_level(node.right, node_level + 1, elem)
    return downlevel

print(get_level(root, 0, 4))
'''Parameters are: root node, root node's level (0) and
the element of the node whose level is to be found'''
```

Binary Trees - Find Height of a Node

If node is None

return -1

else

return 1 + max(left_subtree_height, right_subtree_height)

Do yourself